Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *CIVIL COMMENTS* | ***Context:*** *Gazipur Peri-Urban Low-Literacy Health Chatbot Users (Bangladesh)*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark assumes clean standard English text throughout [Q42, Q56], with comment length the only documented formal variation [Q62, Q64]. The deployment input register is systematically degraded — Roman-script Bangla, phonetic ASCII, absent punctuation, idiosyncratic spacing, code-switching (region YAML writing_systems and input_form_profile; elicitation A3). Dataset sampling confirms 100% standard Latin-script English with only minor informal-spelling variation (DATASET-D12 'straigtup...bullllshit'). The benchmark documents no preprocessing pipeline for non-Latin script, code-switched, or orthographically degraded input, creating a near-complete signal-distribution mismatch.

ID	Evidence
Q42	"The examples are simple sentences that should be clearly toxic or clearly non-toxic, regardless of identity terms present." (p.6)
Q56	"Using these rating techniques we created a dataset of 1.8 million comments, sourced from online comment forums, containing labels for toxicity and identity." (p.7)
Q62	"Table 6 compares results for both TOXICITY@1 and TOXICITY@6 on all metrics, for both short comments (less than 100 characters) and all comments." (p.8)
Q64	"The identities shown in Table 6 are all identities that contained more than 100 examples of short comments." (p.8)
D12	This is straigtup SJW bullllshit!!!! :(— US political internet argot ("SJW"), informal spelling, emoticon

Strengths

- The benchmark explicitly recognizes that comment length affects bias measurement and stratifies short comments separately [Q62, Q64]; this is a structurally transferable lesson — a Gazipur deployment could analogously stratify by input degradation level (e.g., script type or code-switch density).

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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Severe mismatch — benchmark signal is clean US English text [Q42, DATASET-D1 through D25]; deployment signal is Banglish/phonetic Bangla/code-switched degraded text (region YAML writing_systems).

IF-2: Bangladesh infrastructure supports text input but the form is fundamentally different: 63.3% household smartphone penetration [WEB-12] with users typing on phonetic-Bangla or Roman keyboards. The benchmark's clean-English assumption does not match this capture distribution.

IF-3: Domain-specific form differences include: (a) Bengali Unicode vs. Latin script, (b) idiosyncratic romanization (no standard Banglish orthography), (c) absent punctuation/spacing, (d) phonetic vowel omission, (e) code-switch boundaries. None addressed in benchmark documentation.

IF-4: External validity is severely violated: classifier scores measured on clean English have unknown and likely unreliable behavior on degraded Banglish input, meaning AUC/AEG metrics computed on benchmark data do not generalize to deployment input [Q35 cautions exactly this case-by-case handling].

ID	Evidence
Q35	"It's important to note that the correct handling of subtle variations in distributions must be decided on case by case basis." (p.5)
Q42	"The examples are simple sentences that should be clearly toxic or clearly non-toxic, regardless of identity terms present." (p.6)
Q56	"Using these rating techniques we created a dataset of 1.8 million comments, sourced from online comment forums, containing labels for toxicity and identity." (p.7)
D1	You're an idiot. — Clear direct insult, English, standard register
D25	I love that people are sending these guys dildos in the mail now. But… if they really think there's a happy ending in this for any of them, I think they're even more deluded than all of the jokes about them assume. — Sexual innuendo in political context
WEB-12	National household smartphone penetration 63.3% (BBS ICT Use Survey 2023) — confirms mobile-keyboard input dominance (https://www.dhakatribune.com/bangladesh/320526/bbs-proportion-of-households-in-bangladesh-with)
WEB-14	GSMA 2023 reports a 40% mobile-internet gender gap in Bangladesh (highest in Asia), heightening female-user input-form variability (https://www.tbsnews.net/features/panorama/why-bangladeshi-women-lag-behind-men-internet-and-mobile-usage-909321)
WEB-20	BanTH demonstrates that Roman-script Bangla is a distinct NLP problem with its own dataset requirement (https://aclanthology.org/2025.findings-naacl.403/)

Information Gaps

- No documentation of any preprocessing pipeline that could bridge the signal gap

Requires Expert Verification

- Local engineering team to specify Banglish normalization pipeline and script-detection heuristics

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Remediation

Gap 1: Benchmark assumes clean English; deployment input is degraded Banglish/code-switched/phonetic with no preprocessing pipeline documented

Recommendation: Build a Banglish/script-detection/normalization preprocessing layer before any classifier inference, and stratify evaluation by input-form degradation level (analogous to the benchmark's short-vs-all comment stratification [Q62, Q64]) so bias-by-input-form interaction can be diagnosed.

ID	Evidence
Q62	"Table 6 compares results for both TOXICITY@1 and TOXICITY@6 on all metrics, for both short comments (less than 100 characters) and all comments." (p.8)
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